

# Integration (Riemann)

math-foundations concept 15

## First, an example

Before the abstract Darboux machinery, here is the punchline on a concrete case. Take  $f(x) = x^2$  on  $[0, 1]$ . Slice  $[0, 1]$  into  $n = 4$  equal subintervals of width  $\Delta x = 1/4$  and form the right-endpoint Riemann sum

$$R_4 = \sum_{i=1}^4 \left(\frac{i}{4}\right)^2 \cdot \frac{1}{4} = \frac{1}{4} \left( \frac{1}{16} + \frac{4}{16} + \frac{9}{16} + \frac{16}{16} \right) = \frac{30}{64} = 0.46875.$$

*Read aloud:* R-sub-4 equals the sum from i equals 1 to 4 of i-over-4 squared, times one-quarter; this works out to thirty over sixty-four, about 0.469. Refining to  $n = 100$  gives  $\approx 0.3383$ ; the limit is  $\int_0^1 x^2 dx = \frac{1}{3}$ , which the antiderivative  $F(x) = x^3/3$  recovers instantly:  $F(1) - F(0) = 1/3$ . Everything below explains *why* that limit exists and *why* the antiderivative shortcut works.

## 1 Motivation

The Riemann integral assigns to a bounded function  $f : [a, b] \rightarrow \mathbb{R}$  a single number  $\int_a^b f$  that represents the signed area between the graph of  $f$  and the  $x$ -axis. The construction is purely metric: chop  $[a, b]$  into finitely many subintervals, approximate  $f$  on each subinterval by its supremum (overestimate) and infimum (underestimate), and demand that the two estimates converge as the chopping becomes finer.

## 2 Partitions and Darboux Sums

**Definition 1** (Partition). A partition of  $[a, b]$  is a finite ordered set  $P = \{x_0, x_1, \dots, x_n\}$  with

$$a = x_0 < x_1 < \dots < x_n = b.$$

*Read aloud:*  $a$  equals  $x$ -zero, less than  $x$ -one, less than dot-dot-dot, less than  $x$ - $n$ , equals  $b$ . Set  $\Delta x_i = x_i - x_{i-1}$  and define the mesh  $\|P\| = \max_i \Delta x_i$ . A partition  $P'$  refines  $P$  if  $P \subseteq P'$ .

**Definition 2** (Upper and lower sums). Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. For a partition  $P$  as above, set

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x), \quad m_i = \inf_{x \in [x_{i-1}, x_i]} f(x).$$

*Read aloud:* big- $M$ -sub- $i$  is the supremum of  $f$  of  $x$  for  $x$  in the  $i$ -th subinterval; little- $m$ -sub- $i$  is the infimum on the same subinterval. The upper Darboux sum and lower Darboux sum are

$$U(f, P) = \sum_{i=1}^n M_i \Delta x_i, \quad L(f, P) = \sum_{i=1}^n m_i \Delta x_i.$$

*Read aloud:*  $U$  of  $f$  comma  $P$  equals the sum from  $i$  equals 1 to  $n$  of big- $M$ -sub- $i$  times delta- $x$ -sub- $i$ ;  $L$  of  $f$  comma  $P$  is the analogous sum with little- $m$ -sub- $i$ .

**Lemma 1.** If  $P'$  refines  $P$ , then  $L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P)$ . Consequently  $L(f, P_1) \leq U(f, P_2)$  for any two partitions  $P_1, P_2$ .

**Definition 3** (Darboux integrability). Define the upper and lower integrals

$$\overline{\int_a^b} f = \inf_P U(f, P), \quad \underline{\int_a^b} f = \sup_P L(f, P).$$

Read aloud: upper integral from  $a$  to  $b$  of  $f$  equals the infimum over partitions  $P$  of  $U$  of  $f$ ,  $P$ ; lower integral equals the supremum over  $P$  of  $L$  of  $f$ ,  $P$ . We say  $f$  is Riemann (Darboux) integrable on  $[a, b]$  if the two coincide; the common value is denoted  $\int_a^b f(x) dx$ .

**Proposition 2** (Cauchy criterion). A bounded  $f$  is integrable on  $[a, b]$  iff for every  $\varepsilon > 0$  there exists a partition  $P$  with  $U(f, P) - L(f, P) < \varepsilon$ .

*Proof.* ( $\Rightarrow$ ) Suppose  $\overline{\int} f = \underline{\int} f =: I$ . By the definitions of inf and sup, pick  $P_1, P_2$  with  $U(f, P_1) < I + \varepsilon/2$  and  $L(f, P_2) > I - \varepsilon/2$ . The common refinement  $P = P_1 \cup P_2$  satisfies  $U(f, P) \leq U(f, P_1)$  and  $L(f, P) \geq L(f, P_2)$ , so  $U(f, P) - L(f, P) < \varepsilon$ .

( $\Leftarrow$ ) For every  $\varepsilon$  we have  $0 \leq \overline{\int} f - \underline{\int} f \leq U(f, P) - L(f, P) < \varepsilon$ . Hence the difference is 0.  $\square$

### 3 Continuity Implies Integrability

**Theorem 3.** Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable.

*Proof.* Because  $[a, b]$  is compact,  $f$  is bounded and uniformly continuous (Heine–Cantor). Fix  $\varepsilon > 0$ . By uniform continuity, choose  $\delta > 0$  such that

$$x, y \in [a, b], |x - y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{b - a}.$$

Read aloud: for all  $x, y$  in the closed interval  $a$  to  $b$ , if the absolute value of  $x$  minus  $y$  is less than  $\delta$ , then the absolute value of  $f$  of  $x$  minus  $f$  of  $y$  is less than  $\varepsilon$  over  $b$  minus  $a$ . Take any partition  $P = \{x_0, \dots, x_n\}$  with mesh  $\|P\| < \delta$ .

On each subinterval  $[x_{i-1}, x_i]$ , the function  $f$  attains its sup  $M_i$  and inf  $m_i$  (continuous on a compact set), say at points  $\xi_i, \eta_i \in [x_{i-1}, x_i]$ . Since  $|\xi_i - \eta_i| \leq \Delta x_i < \delta$ ,

$$M_i - m_i = f(\xi_i) - f(\eta_i) < \frac{\varepsilon}{b - a}.$$

Read aloud: big- $M$ -sub- $i$  minus little- $m$ -sub- $i$  equals  $f$  of  $\xi$ -sub- $i$  minus  $f$  of  $\eta$ -sub- $i$ , which is less than  $\varepsilon$  over  $b$  minus  $a$ .

Summing over  $i$ ,

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i - m_i) \Delta x_i < \frac{\varepsilon}{b - a} \sum_{i=1}^n \Delta x_i = \frac{\varepsilon}{b - a} \cdot (b - a) = \varepsilon.$$

Read aloud:  $U$  of  $f$ ,  $P$  minus  $L$  of  $f$ ,  $P$  equals the sum from  $i$  equals 1 to  $n$  of  $M$ -sub- $i$  minus  $m$ -sub- $i$ , times  $\Delta x$ -sub- $i$ , which is less than  $\varepsilon$  over  $b$  minus  $a$ , times the total length  $b$  minus  $a$ , equals  $\varepsilon$ .

By Proposition 2,  $f$  is integrable.  $\square$

## 4 The Fundamental Theorem of Calculus

**Theorem 4** (FTC, Part 1). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, and define  $F : [a, b] \rightarrow \mathbb{R}$  by  $F(x) = \int_a^x f(t) dt$ . Then  $F$  is differentiable on  $(a, b)$  and  $F'(x) = f(x)$ .*

*Sketch.* For  $h \neq 0$ ,  $\frac{F(x+h)-F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt$ . Bound the integrand by  $f(x) \pm \varepsilon$  using continuity of  $f$  at  $x$ , and let  $h \rightarrow 0$ .  $\square$

**Theorem 5** (FTC, Part 2). *If  $f$  is continuous on  $[a, b]$  and  $G$  is any antiderivative of  $f$ , then  $\int_a^b f(t) dt = G(b) - G(a)$ .*

*Sketch.* Both  $F$  from Part 1 and  $G$  are antiderivatives of  $f$ , so they differ by a constant:  $G(x) = F(x) + C$ . Then  $G(b) - G(a) = F(b) - F(a) = \int_a^b f - 0$ .  $\square$

## 5 Worked Example

For  $f(x) = x^2$  on  $[0, 1]$  with equal partition  $x_i = i/n$ ,

$$U(f, P_n) = \sum_{i=1}^n \frac{i^2}{n^2} \cdot \frac{1}{n} = \frac{(n+1)(2n+1)}{6n^2} \xrightarrow{n \rightarrow \infty} \frac{1}{3}.$$

*Read aloud:* U of f, P-sub-n equals the sum from i equals 1 to n of i-squared over n-squared, times one over n; this simplifies to (n plus 1)(2n plus 1) over 6 n-squared, which tends to one-third as n goes to infinity. By FTC2 the antiderivative  $x^3/3$  also yields  $1/3 - 0 = 1/3$ .